FREE RESOURCE — UNIVERSITIES & RESEARCH

UAS Research Program Operator Training Framework

Bridging the gap between Part 107 certification and research-capable UAS operation. For faculty and program directors.

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About This Framework

For faculty principal investigators, aviation program directors, and lab managers who need to structure operator training for their research programs. Covers the capability gap between regulatory certification and research-ready operation, and provides a structured pathway to close it.

The Part 107 Gap

Part 107 Certifies	Research Operations Require
Written knowledge exam (airspace, weather, regulations)	Independent platform operation without support staff
Aeronautical decision-making concepts — no practical test	Field repair and pre-flight troubleshooting capability
No platform-specific training required	Platform-specific configuration and failure analysis
No proficiency evaluation — exam only	Documented proficiency defensible to IRB and institution
No training on autonomous systems or edge computing	Operator-supervised autonomy for AI-enabled research platforms

Sample Training Pathway for Research Assistants

Recommended baseline for graduate research assistants who will operate research platforms independently.

Phase	Requirement	FFA Course	Outcome

Phase 1 Foundation	FAA Part 107 certification	FFA-201 prep or self-study	Regulatory compliance baseline
Phase 2 Platform Ops	Platform-specific operations and failure analysis	FFF-401 or FFA-401	Independent operation, field repair
Phase 3 Specialization	Domain-specific capability (RF, electronics, AI)	FFR-101, FFF-201, or FFA-401	Research-specific platform proficiency
Phase 4 Documentation	Capability assessment for IRB and program records	Assessment Services	1–5 ratings across 5 domains, defensible to institution

Course Integration Guide

How FFA courses map to university research program requirements. Academic pricing available on all courses.

FFA-201

Autonomy Fundamentals for Operators

Foundational autonomy concepts for operators entering AI-enabled research. No prior ML or CS background required. Applicable to any lab deploying autonomous platforms.

2-day course | Academic pricing: $3,000/student

FFA-401

Operational Edge Computing for UAS

Deploy real-time AI on Jetson Orin Nano — one per student, students keep hardware. Designed for robotics, aerospace, and CS research labs. No ML background required.

5-day course | Academic pricing: $4,000/student | Hardware included & retained

FFF-201

Deployed Electronics Repair & Diagnostics

Component-level repair, soldering, and failure analysis. Builds lab capability to sustain research platforms without external vendor support.

5-day course | Academic pricing: $5,500/student

FFF-301

Additive Manufacturing for Sustainment

3D printing and rapid prototyping for platform repair and modification. Applicable to any lab that runs its own platforms in the field.

3-day course | Academic pricing: $3,000/student

FFR-101

RF Fundamentals & Spectrum Awareness

Baseline RF literacy for research operators. Understand link behavior, interference, and communications resilience without an engineering background.

2-day course | Academic pricing: $3,000/student

Sample Flight Test Protocol Checklist

Adapt for your platform and IRB requirements.

PRE-MISSION (DAY BEFORE)

- [ ] Airspace authorization obtained (LAANC or waiver as required)
- [ ] Weather briefing reviewed — wind, precipitation, visibility within limits
- [ ] Platform inspected — motors, props, battery, frame integrity

- [ ] Emergency procedures briefed to all personnel on site

DAY-OF PRE-FLIGHT

- [ ] Battery voltage and health verified
- [ ] Flight controller calibration confirmed (compass, IMU)
- [ ] RC link and telemetry tested at ground level
- [ ] GPS satellite acquisition verified (minimum 8 satellites)
- [ ] Research payload power-up and data logging confirmed

POST-FLIGHT

- [ ] Flight log downloaded and archived
- [ ] Battery discharged to storage voltage if not flying within 24 hours
- [ ] Any anomalies documented in maintenance log

Ready to Train Your Research Team?

Tell us about your program, platforms, and timeline. We build a custom training package and return a proposal within 48 hours. Academic pricing available on all 10 courses.

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